FIELD / 2458

A small laboratory, built as an entity-component-system

Tair Asset | AIB-2401 | Student ID 242458 | PA4

Six primitive-built exhibits occupy a 4 m ring. ID digits 8, 5, 4, 2 select science lab, pulse, 1.5-second gaze fuse and ring layout. Core A-Frame 1.8.0 supplies Three.js r184; three original CC0 textures preload through a-assets. The three custom components and gallery system use the x2458- prefix.

E1 / A component through time

Prediction. init runs once; the first update has no previous configuration. Changing amplitude calls update again. Pause suppresses motion; play resumes it. Removing the component calls remove; subsequently deleting its entity cannot remove that same component twice.

Timestamp, ms	Observed method	Previous configuration
6135.3	init	-
6136.1	update	{}
6136.6	play	-
6138.8	tick	-
6735.7	update	{"type":"pulse","amplitude":0.2,"period":4,"phase":0}
6736.4	pause	-
6736.9	play	-
7171.2	tick	-
7267.7	pause	-
7269.4	remove	-

Result and explanation. The log follows the prediction. oldData is empty on initialization and contains the previous parsed values after setAttribute. Motion keeps base transforms and an accumulated phase; period changes affect future phase advance. Amplitude/type changes preserve the current pose through a decaying offset. Local time stops during pause. Tick logging is sampled to avoid flooding the console.

E2 / Rechecking the chapter

Prediction. Reusing the chapter loader will recreate runtime objects on a configuration change. Calling the asset manager a system does not make it an AFRAME.registerSystem instance.

Result. A test fixture reconstructed from supplied Ch. 8 pp. 12-15 replaced both texture and material after changing src; the previous texture was not disposed. Before and after, this.data contained Texture, MeshLambertMaterial and Mesh objects. In the corrected loader, this.data contains configuration only; objects live on the instance. The actual system list contains x2458-gallery but no a-assets.

Explanation. Parsed, serializable attributes describe configuration; mesh/material/texture ownership needs explicit lifecycle cleanup. A-assets is an AAssets custom element extending ANode, with asset-loading behaviour. It is an asset-management subsystem, not the scene singleton created by registerSystem.

E3 / A floor at a grazing angle

Prediction. Nearest minification will alias the repeated grid. Mipmaps blend distant detail; anisotropy should preserve oblique detail better.

Setting	Minification	Magnification	Anisotropy
Chapter	NearestFilter	LinearFilter remains	16
Three.js defaults	LinearMipmapLinearFilter	LinearFilter	1
Chosen	LinearMipmapLinearFilter	LinearFilter	8

Result. The same fixed camera and paused scene produced the three attached images. Nearest gives harder, less stable distant grid detail; default mipmaps soften it; the chosen setting balances smoother distance transitions with oblique detail. Static images show spatial aliasing, not temporal shimmer by themselves.

Explanation. Minification handles a texel footprint smaller than a screen pixel; magnification enlarges texels. Mipmaps provide prefiltered smaller images. The chapter writes maxFilter, which is absent from a fresh Three.js Texture. The valid property is magFilter; assigning maxFilter merely adds an unused JavaScript field. The corrected loader uses magFilter and clamps anisotropy to the device limit. Ground repetition is 16 x 16.

E4 / What the scene builds

Prediction. A-scene will create Three.js rendering objects and manage session setup that PA3 handled explicitly.

Console inspection	Observed result
sceneEl.renderer.isWebGLRenderer	True
sceneEl.object3D.type	Scene
sceneEl.camera.type	PerspectiveCamera
sceneEl.renderer.xr	Present

Explanation. In A-Frame v1.8.0 src/core/scene/a-scene.js, enterVR starts at line 295; lines 320-329 request the session and pass it to renderer.xr.setSession. PA3 explicitly requested immersive-ar, negotiated features, obtained reference spaces and cleaned up hit testing. A-Frame wraps common scene/session work. A company building a specialised CAD renderer may prefer raw Three.js; a small interactive exhibition benefits from A-Frame composition and reusable components.

Measurement context

The desktop experiments use the actual browser and Intel UHD graphics reported by WebGL. Tests are automated, not physical-headset measurements. The VR check uses official Immersive Web Emulator 2.0 with a Meta Quest 3 profile; renderer.xr.isPresenting and a gaze selection were both observed. No physical headset comfort or 72 Hz claim is made.

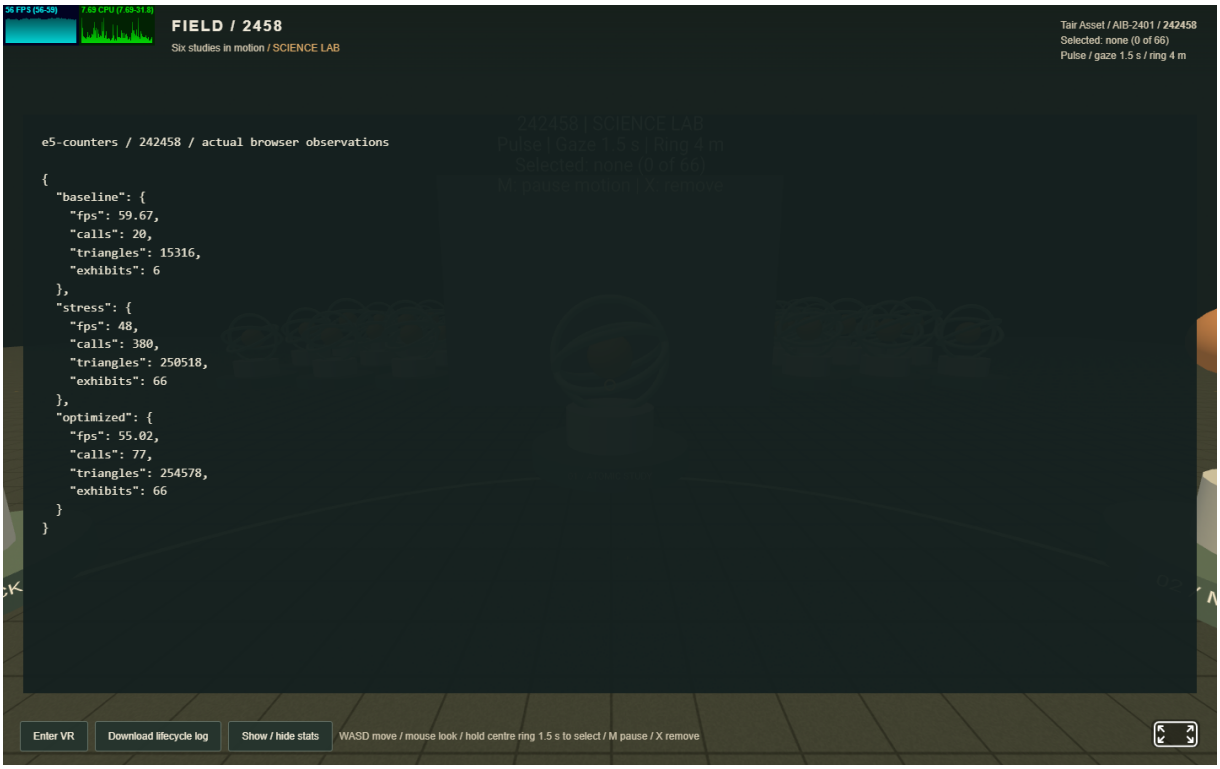
E5 / Rendering more exhibits

Prediction. Sixty added exhibit entities will increase draw calls and triangle work; instancing compatible meshes should reduce submissions without removing exhibits.

Scene	Exhibits	FPS	Draw calls	Triangles
Original	6	59.67	20	15316
+60 exhibits	66	48	380	250518
Instanced	66	55.02	77	254578

Result. Each value is a three-second desktop measurement with stats enabled. Compatible geometry/material groups become InstancedMesh objects, while the exhibit entities remain registered. Additional instances can render outside the original per-object frustum culling, so triangles need not fall with draw calls.

Explanation. Draw calls incur CPU/driver submission and state-change costs. VR must supply two views within a tighter frame budget, making unnecessary submissions costly. Instancing reduces calls rather than pretending to reduce model complexity. FPS is a sampled observation, not a guarantee for every machine.



E5: actual browser counters displayed by the experiment harness; ID remains visible.

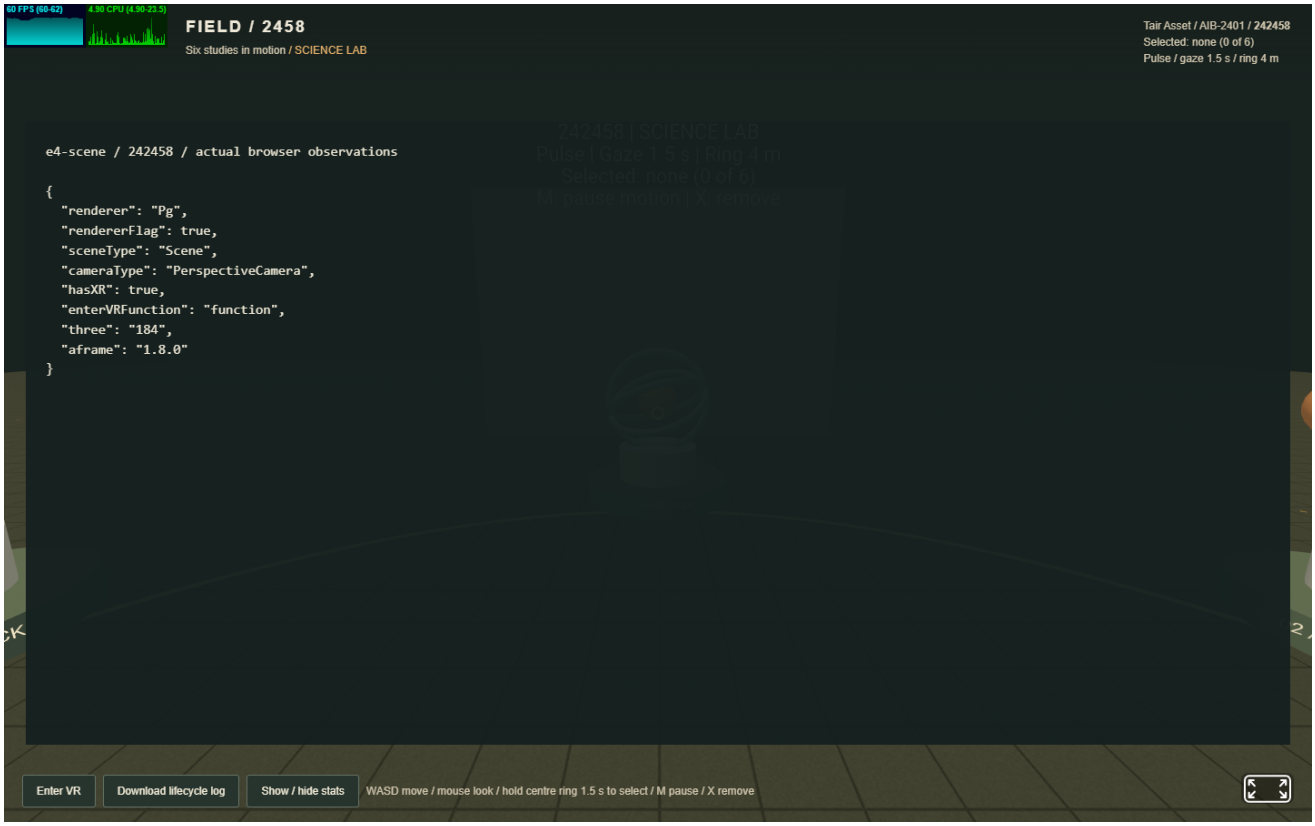
E6 / Comfort and design

The 4 m ring leaves an open centre; neighbouring exhibit centres are 4 m apart. The camera starts at 1.6 m. The status text is 4.4 m wide by configuration; its measured line angle is approximately 2.21 degrees at the starting view, using 2 atan(line height / (2 distance)). Exhibit labels face the centre. A visible gaze ring and 1.5-second fuse reduce accidental selection; selected text and amplitude give feedback. Pulse amplitude is modest rather than moving exhibits towards the visitor. Smooth WASD travel can cause discomfort in a headset, so the gallery removes WASD on entering VR and restores it on exit. VR viewing remains stationary with head turns. These are design choices; no physical-headset comfort study was performed.

Development log

The gaze ring was present but world geometry could paint over it.
A screenshot and mesh inspection exposed the render-order issue.
Commit ade0824 draws the ring last and freezes resumed timing.
Child-event regression checks then found double selection; dad2353 stops the original propagation.

E4 / Browser inspection



Actual renderer, scene and camera observations from the running application.

Observed lifecycle and chapter data



E1: timestamped hook calls for the required sequence. The final entity deletion follows removal of its motion component.



E2: before/after runtime inspection of the chapter fixture, rendered by the test harness.

Three settings, one camera



E3 / Chapter: NearestFilter, anisotropy 16; maxFilter does not change magnification.

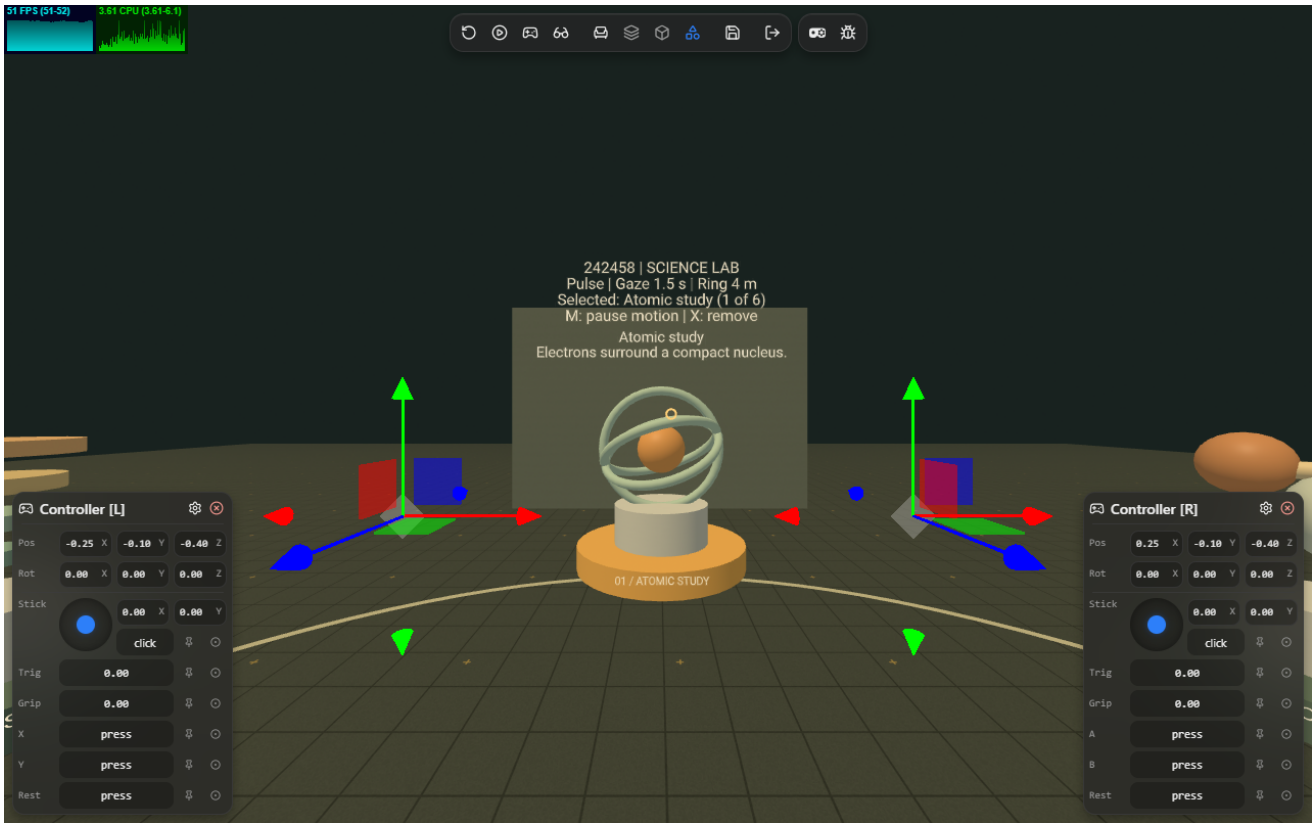


E3 / Three.js defaults: trilinear mipmap minification, linear magnification, anisotropy 1.

Chosen filtering and VR selection



E3 / Chosen: trilinear mipmaps, linear magnification, supported anisotropy up to 8.



VR / Official emulator, active WebXR session, gaze-selected Atomic study. This is emulation, not a physical headset recording.

Sources: supplied PA4 assignment; supplied Baruah Ch. 8 PDF pp. 6, 12-15, 18 and corresponding presentation; A-Frame v1.8.0 a-assets.js and a-scene.js inspected for E2/E4. Raw measurements and captures are retained locally.